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Do bottom-up parsing for this grammar:

$S' \rightarrow S$
 $S \rightarrow C C$
 $C \rightarrow c C$
 $C \rightarrow d$

Use these string to practice

- $c d d$
- $c c d d$
- $d d$

Answer:

S' = start,
 c, d = terminal
 C = non-terminal

1. Parsing String $c d d$

Stack	Input	Action
\$	$c d d \$$	shift
$\$ c$	$d d \$$	shift
$\$ c d$	$d \$$	reduce $C \rightarrow d$
$\$ c C$	$d \$$	reduce $C \rightarrow c C$
$\$ C$	$d \$$	shift
$\$ C d$	$\$$	reduce $C \rightarrow d$
$\$ C C$	$\$$	reduce $S \rightarrow C C$
$\$ S$	$\$$	reduce $S' \rightarrow S$
$\$ S'$	$\$$	accept

2. Parsing String c c d d

Stack	Input	Action
\$	c c d d \$	shift
\$ c	c d d \$	shift
\$ c c	d d \$	shift
\$ c c d	d \$	reduce $C \rightarrow d$
\$ c c C	d \$	reduce $C \rightarrow c C$
\$ c C	d \$	shift
\$ c C d	\$	reduce $C \rightarrow d$
\$ c C C	\$	reduce $C \rightarrow c C$
\$ C C	\$	reduce $S \rightarrow C C$
\$ S	\$	reduce $S' \rightarrow S$
\$ S'	\$	accept

3. Parsing String d d

Stack	Input	Action
\$	d d \$	shift
\$ d	d \$	reduce $C \rightarrow d$
\$ C	d \$	shift
\$ C d	\$	reduce $C \rightarrow d$
\$ C C	\$	reduce $S \rightarrow C C$
\$ S	\$	reduce $S' \rightarrow S$
\$ S	\$	accept